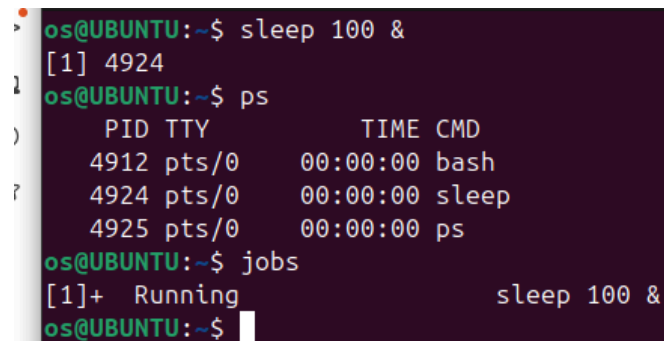


Week 2 — Processes & Signals

Assignment 1: Process Investigation

1. Start multiple background processes:

- `sleep`



```
os@UBUNTU:~$ sleep 100 &
[1] 4924
os@UBUNTU:~$ ps
  PID TTY          TIME CMD
 4912 pts/0    00:00:00 bash
 4924 pts/0    00:00:00 sleep
 4925 pts/0    00:00:00 ps
os@UBUNTU:~$ jobs
[1]+  Running                  sleep 100 &
os@UBUNTU:~$
```

- `yes > /dev/null`

2. Use:

- `ps`

```

os@UBUNTU:~$ sleep 100 &
[1] 4924
os@UBUNTU:~$ ps
  PID TTY          TIME CMD
  4912 pts/0        00:00:00 bash
  4924 pts/0        00:00:00 sleep
  4925 pts/0        00:00:00 ps
os@UBUNTU:~$ jobs
[1]+  Running                  sleep 100 &
os@UBUNTU:~$ ps
  PID TTY          TIME CMD
  4912 pts/0        00:00:00 bash
  4924 pts/0        00:00:00 sleep
  4943 pts/0        00:00:00 ps
os@UBUNTU:~$

```

o top

```

os@UBUNTU:~$ ps
  PID TTY          TIME CMD
  4912 pts/0        00:00:00 bash
  4924 pts/0        00:00:00 sleep
  4925 pts/0        00:00:00 ps
os@UBUNTU:~$ jobs
[1]+  Running                  sleep 100 &
os@UBUNTU:~$ ps
  PID TTY          TIME CMD
  4912 pts/0        00:00:00 bash
  4924 pts/0        00:00:00 sleep
  4943 pts/0        00:00:00 ps
os@UBUNTU:~$ top

```

top - 16:53:22 up 1:24, 1 user, load average: 1.63, 0.89, 0.57
Tasks: 206 total, 1 running, 205 sleeping, 0 stopped, 0 zombie
%Cpu(s): 32.7 us, 24.8 sy, 0.0 ni, 39.4 id, 0.0 wa, 0.0 hi, 3.1 si, 0.0 st
MiB Mem : 5924.8 total, 333.3 free, 3066.1 used, 2356.0 buff/cache
MiB Swap: 0.0 total, 0.0 free, 0.0 used. 2858.7 avail Mem

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
1875	os	20	0	4120420	502588	180228	S	104.0	8.3	19:55.29	gnome-s+
2682	os	20	0	11.6g	687908	291056	S	41.7	11.3	3:32.53	firefox
3729	os	20	0	3077476	452172	148272	S	9.9	7.5	0:19.62	Isolate+
4427	root	20	0	0	0	0	I	0.7	0.0	0:00.89	kworker+
4905	os	20	0	553852	54124	43124	S	0.7	0.9	0:01.00	gnome-t

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Q

Explain:

-

Process hierarchy is can be saw with the help of pstree. This show for what process a new process is created and what process is actual parent

-

With the help of ps aux we can see which process is taking how much CPU

```

> ofilo: command not found
[1]-  Done                sleep 100
os@UBUNTU:~$ ps aux
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root         1  0.0  0.2 23280 14696 ?        Ss   15:28   0:03 /sbin/init sp
root         2  0.0  0.0      0     0 ?        S    15:28   0:00 [kthreadd]
root         3  0.0  0.0      0     0 ?        S    15:28   0:00 [pool_workque
root         4  0.0  0.0      0     0 ?        I<   15:28   0:00 [kworker/R-rc
root         5  0.0  0.0      0     0 ?        I<   15:28   0:00 [kworker/R-sy
root         6  0.0  0.0      0     0 ?        I<   15:28   0:00 [kworker/R-kv
root         7  0.0  0.0      0     0 ?        I<   15:28   0:00 [kworker/R-sl
root         8  0.0  0.0      0     0 ?        I<   15:28   0:00 [kworker/R-ne
root        12  0.0  0.0      0     0 ?        I    15:28   0:00 [kworker/u8:0
root        13  0.0  0.0      0     0 ?        I<   15:28   0:00 [kworker/R-mm
root        14  0.0  0.0      0     0 ?        S    15:28   0:00 [ksoftirqd/0]
root        15  0.0  0.0      0     0 ?        I    15:28   0:03 [rcu_preempt]
root        16  0.0  0.0      0     0 ?        S    15:28   0:00 [rcu_exp_par_
root        17  0.0  0.0      0     0 ?        S    15:28   0:00 [rcu_exp_gp_k
root        18  0.0  0.0      0     0 ?        S    15:28   0:00 [migration/0]
root        19  0.0  0.0      0     0 ?        S    15:28   0:00 [idle_inject/
root        20  0.0  0.0      0     0 ?        S    15:28   0:00 [cpuhp/0]
root        21  0.0  0.0      0     0 ?        S    15:28   0:00 [cpuhp/1]
root        22  0.0  0.0      0     0 ?        S    15:28   0:00 [idle_inject/
root        23  0.0  0.0      0     0 ?        S    15:28   0:00 [migration/1]

```

- Open file descriptors

Lsof

Assignment 2: Signal Lab

1. Start a long-running process.
2. Send:
 - **SIGINT**

```
os@UBUNTU:~$ trap "echo Stop" SIGINT
os@UBUNTU:~$ sleep 100
^C
os@UBUNTU:~$ sleep 100

^C
os@UBUNTU:~$
```

-

- SIGSTOP

```
os@UBUNTU:~$ sleep 100 &
[1] 4495
os@UBUNTU:~$ kill -STOP 4495
os@UBUNTU:~$ jobs
[1]+  Stopped                  sleep 100
os@UBUNTU:~$
```

- SIGCONT

```

[1]+  Stopped                  sleep 100
os@UBUNTU:~$ kill -CONT 4495
os@UBUNTU:~$ JOBS
JOBS: command not found
os@UBUNTU:~$ jobs
[1]+  Running                  sleep 100 &
os@UBUNTU:~$

```

- SIGKILL

```

[1]+  Running                  sleep 100 &
os@UBUNTU:~$ kill -9 4495
bash: kill: (4495) - No such process
[1]+  Done                    sleep 100
os@UBUNTU:~$ jobs
os@UBUNTU:~$

```

Document behavioral differences.

Assignment 3: PATH Hijack Experiment

1. Create a fake command in a custom directory.
2. Modify `PATH`.
3. Override system command behavior.

Explain why this is dangerous.

